

Part I — Foundations of Physics · Chapter 5

Graphs, Models, and Experimental Uncertainty

Reading physical meaning from data, building simplified representations of reality, and interpreting measurements honestly

A table of numbers from an experiment tells you what was measured. A graph of those numbers reveals the relationship. A model tells you why. And an uncertainty estimate tells you how much to trust it all. These four tools — graph, table, model, uncertainty — are how physics converts raw data into knowledge. This chapter builds the analytical habits you will use in every experiment and every data-driven problem for the rest of this book.

5.1 Why Physicists Use Graphs and Models

A physicist rarely stops at measuring. The measurement is the beginning, not the end. Once data is collected, the physicist asks: What pattern do these numbers reveal? What relationship connects the variables? Does that relationship match what theory predicts? Could the agreement be accidental? These questions require tools beyond arithmetic.

Graphs are among the most powerful of those tools. A well-constructed graph makes a pattern visible at a glance that would take minutes to detect by scanning a data table. A curve's shape tells you immediately whether a relationship is linear, quadratic, inverse, or something else. The slope of a line on a graph can represent a physical quantity — velocity on a position-time graph, acceleration on a velocity-time graph, spring constant on a force-extension graph. The area under a curve on a graph can represent another physical quantity — distance on a velocity-time graph, work done on a force-displacement graph.

Models are simplifications. No experiment perfectly matches the idealized equations of a textbook, because the real world has air resistance, friction, temperature fluctuations, and imperfect instruments. A model selects the most important factors, ignores the less important ones, and produces a mathematical description that is tractable without being false. Learning to work with models — to know when they apply and when they fail — is one of the most important intellectual skills in physics.

"All models are wrong, but some are useful." — George Box, statistician

5.2 Independent and Dependent Variables

Every experiment involves at least two variables: the quantity the experimenter controls or varies, and the quantity that responds. Understanding which is which is essential before constructing a table or graph.

The independent variable is the quantity the experimenter deliberately changes or controls. It is the input to the experiment. The dependent variable is the quantity that responds to those changes; it is the output or result. In a well-designed experiment, all other relevant quantities (called controlled variables or controlled conditions) are held constant while the independent variable is varied, so that any change in the dependent variable can be attributed to the independent variable alone.

Experiment	Independent Variable	Dependent Variable	Controlled Variables
Dropping balls from different heights	Height of drop	Time to fall	Ball size, air, location
Stretching a spring with different weights	Mass hanging (weight)	Extension of spring	Spring type, temperature
Heating water with a burner	Time of heating	Temperature of water	Mass of water, flame size
Charging a capacitor	Time	Voltage across capacitor	Capacitance, voltage source

Table 5.1 Examples of independent and dependent variables in physics experiments.

The Rule: Convention: The independent variable is plotted on the horizontal axis (x-axis). The dependent variable is plotted on the vertical axis (y-axis). This is the standard scientific convention and should be followed in all your graphs.

5.3 Reading Scientific Graphs

A well-constructed scientific graph contains several standard elements. Learning to read each element quickly is a skill used in every chapter of this book and in every science course you will take.

Graph Element	Purpose	What to Look For
---------------	---------	------------------

Title	Identifies what the graph shows	Should state dependent vs. independent variable
Horizontal axis label	Names the independent variable	Include variable name and unit in parentheses
Vertical axis label	Names the dependent variable	Include variable name and unit in parentheses
Scale markings	Provide numerical values	Check that scale is uniform and zero is shown if relevant
Data points	Show actual measured values	Individual points, often with error bars
Best-fit line/curve	Shows the trend in the data	Passes among data points, not necessarily through all
Error bars	Show measurement uncertainty	Vertical bars show uncertainty in the y-variable
Legend	Identifies multiple data sets	Used when more than one set of data is plotted

Table 5.2 *Standard elements of a scientific graph and their meaning.*

When reading a graph, extract specific values by reading coordinates at points of interest. Estimate intermediate values by interpolation (reading between data points). Avoid extrapolation (reading beyond the range of the data) unless you have strong theoretical reasons to believe the relationship continues.

Common Mistake: Extrapolating beyond the range of your data is unreliable. A linear relationship observed between 0 and 10 s may become nonlinear at 20 s. Always state clearly when an answer depends on extrapolation.

5.4 Constructing Proper Graphs

Constructing a graph is not merely drawing data points on paper. A scientific graph must be built to communicate clearly and honestly.

Steps for Constructing a Scientific Graph

Step 1: Identify which variable is independent (x-axis) and which is dependent (y-axis).

Step 2: Choose a scale that spreads the data across at least 75% of the available graph area. Scales that bunch all data into one corner waste information.

Step 3: Label both axes with variable name and unit in parentheses: Time (s), Velocity (m/s), Force (N).

Step 4: Plot data points as small, clear symbols (dots, crosses, or circles). Do not connect points with zigzag lines.

Step 5: Draw a best-fit line or curve that follows the trend of the data. It need not pass through every point, and need not pass through the origin unless theory requires it.

Step 6: Add a title that identifies the graph: "Velocity vs. Time for Uniformly Accelerating Car."

Tip: The best-fit line should pass among the data points, with roughly equal numbers of points above and below. Do not force the line through the origin unless the theory specifically requires zero-zero. Use a ruler for straight lines; use a smooth curve for nonlinear data.

Common Mistake: Do not connect data points with straight line segments between each pair. This creates a zigzag pattern that overfits the noise in the data. Draw a single smooth best-fit curve instead.

5.5 Interpreting Slope and Rate of Change

The slope of a line on a graph tells you how rapidly the dependent variable changes with respect to the independent variable. Slope is one of the most important physical quantities you can read from a graph.

$$\text{slope} = \text{rise} / \text{run} = \Delta y / \Delta x = (y_2 - y_1) / (x_2 - x_1)$$

Choose two widely separated points on the best-fit line — not individual data points — to calculate slope.

The physical meaning of the slope depends on what is plotted. On a position-time graph, slope is velocity (m/s). On a velocity-time graph, slope is acceleration (m/s²). On a force-extension graph for a spring, slope is the spring constant k (N/m). The slope always has units equal to the y-units divided by the x-units.

Graph Type (y vs. x)	Slope Units	Physical Meaning of Slope
Position (m) vs. Time (s)	m/s	Velocity
Velocity (m/s) vs. Time (s)	m/s ²	Acceleration
Force (N) vs. Extension (m)	N/m	Spring constant k

Voltage (V) vs. Current (A)	$V/A = \Omega$	Resistance R (Ohm's law)
Temperature (°C) vs. Time (s)	K/s or °C/s	Rate of temperature change
Work (J) vs. Displacement (m)	$J/m = N$	Force (if constant)

Table 5.3 *Physical meaning of slope for common graph types in introductory physics.*

Common Mistake: Calculate slope using two points on the best-fit line, not two raw data points. Data points scatter due to experimental error; the best-fit line averages through that scatter. Using raw data points to compute slope imports the measurement error directly into your calculation of slope.

A negative slope indicates that the dependent variable decreases as the independent variable increases. A slope of zero means the dependent variable is constant. These physical interpretations are as important as the numerical value.

5.6 Interpreting Area Under a Curve

While slope gives a rate of change, the area enclosed between a graph and the horizontal axis gives an accumulated quantity. This is the graphical version of integration — a concept from calculus — but it can be understood geometrically without calculus.

Graph Type (y vs. x)	Area Units	Physical Meaning of Area
Velocity (m/s) vs. Time (s)	$m/s \times s = m$	Displacement
Acceleration (m/s ²) vs. Time (s)	m/s	Change in velocity
Force (N) vs. Displacement (m)	$N \times m = J$	Work done
Power (W) vs. Time (s)	$W \times s = J$	Energy transferred
Current (A) vs. Time (s)	$A \times s = C$	Electric charge transferred

Table 5.4 *Physical meaning of the area under the graph for common graph types.*

To estimate the area under a curve when the relationship is not simple, count grid squares under the curve, or break the region into triangles and rectangles and add

their areas. The units of area are always (y-axis units) $\times$ (x-axis units).

Example — Area under a velocity-time graph A car accelerates from 0 to 20 m/s in 5.0 s. On a v-t graph, this is a right triangle with base 5.0 s and height 20 m/s. Area = $\frac{1}{2} \times 5.0 \text{ s} \times 20 \text{ m/s} = 50 \text{ m}$. The car traveled 50 m during the acceleration.

Bridge Note *Slope and area under a curve are the graphical versions of differentiation and integration. In the advanced series (Volume I and beyond), you will compute these analytically using calculus. The graphical intuition developed here makes those formal operations much more meaningful.*

5.7 Linear vs. Nonlinear Relationships

Not all physical relationships produce straight lines on a standard graph. Recognizing the shape of a curve gives you important information about the mathematical relationship between the variables.

Curve Shape	Mathematical Form	Physical Example
Straight line through origin	$y = kx$ (direct proportion)	Hooke's law: $F = kx$; Ohm's law: $V = IR$
Straight line, non-zero intercept	$y = mx + b$	Position with initial velocity: $x = x_0 + vt$
Upward-curving parabola	$y = kx^2$	Distance in free fall: $d = \frac{1}{2}gt^2$
Downward-curving (square root)	$y = k\sqrt{x}$	Period of pendulum: $T = 2\pi\sqrt{L/g}$
Hyperbola (inverse proportion)	$y = k/x$	Boyle's law: $P = k/V$
Exponential growth	$y = ae^{(kx)}$	Bacterial growth; RC circuit charging (early)
Exponential decay	$y = ae^{(-kx)}$	Radioactive decay; RC circuit discharge

Table 5.5 Common curve shapes in physics and their mathematical forms.

Linearizing a Nonlinear Relationship

A powerful technique is to replot the data in a way that produces a straight line even when the original relationship is nonlinear. This is called linearization. A straight line is

easier to fit precisely and its slope and intercept are easy to extract.

Example — Linearizing $y = kx^2$ If you plot y vs. x and get a parabola, try plotting y vs. x^2 . If the relationship is truly $y = kx^2$, the new graph will be a straight line through the origin with slope k .

Example — Linearizing $y = k/x$ If y vs. x gives a hyperbola, plot y vs. $1/x$. If truly $y = k/x$, the result is a straight line through the origin with slope k .

Tip: Whenever a graph produces a curve, ask: what power of x , or what function of x , would make this a straight line? Linearization converts curve-fitting — which requires advanced methods — into simple slope-reading.

5.8 Physical Models and Idealizations

A physical model is a simplified representation of a real system that captures the essential physics while deliberately ignoring secondary complications. Models are not failures of precision; they are strategic decisions about which factors dominate.

Every model makes idealizations — deliberate simplifications that make the mathematics tractable without making the physics fundamentally wrong. A model is valid when the ignored factors produce corrections smaller than the precision of the measurement. It fails when those corrections become significant.

Idealization	What It Ignores	When It Is Valid	When It Fails
Frictionless surface	Friction	When μ is very small	High-friction situations; precision experiments
Point mass	Size and shape	When object size $\ll$ length scale of motion	Rotation; large extended objects
Massless string	String mass	String mass $\ll$ hanging mass	Heavy cables; long ropes under tension
Ideal gas	Intermolecular forces; finite particle size	At low pressure, high temperature	Near phase transitions; high-pressure conditions
Uniform gravitational field	Variation of g with altitude	Within a few km of Earth's surface	Satellite orbits; long-range ballistics

No air resistance	Drag force	Low speed; dense objects	High speed; light/large objects
-------------------	------------	--------------------------	---------------------------------

Table 5.6 *Common physical idealizations, what they ignore, and when they are and are not valid.*

Recognizing which model applies to a situation is a judgment call that improves with experience. A useful test: estimate the size of the neglected effect. If it is 10 times smaller than the measurement uncertainty, ignoring it is justified. If it is comparable to or larger than the uncertainty, the model is inadequate and must be refined.

Historical Note: Newton's original laws, applied with the assumption of no air resistance, predict that a cannonball's range is maximized at 45° . This model works beautifully for small, dense projectiles at moderate speeds. For lightweight shells at high velocity, where air resistance is significant, the optimal angle drops to around 30° – 35° . The model failed not because Newton's laws are wrong, but because the air-resistance idealization broke down.

5.9 Assumptions in Scientific Modeling

Every model rests on explicit and implicit assumptions. Making assumptions explicit — stating them clearly before doing the calculation — is a hallmark of careful scientific reasoning. It tells both you and your reader exactly when the model can be trusted.

How to State Assumptions

- Identify what quantities are being held constant that might, in reality, vary (e.g., "we assume the spring constant does not change with temperature").
- Identify forces or effects being neglected (e.g., "we neglect air resistance").
- Identify the range of conditions within which the model is expected to be valid (e.g., "this model applies for speeds much less than the speed of light").
- Identify whether the system is open or closed, isolated or in contact with an environment (e.g., "we assume no energy enters or leaves the system").

Testing Whether an Assumption Is Justified

An assumption is justified if the quantity being neglected produces a correction smaller than the measurement uncertainty. Estimate the correction. Compare it to the uncertainty. If the correction is negligible, the assumption is justified for this measurement. If not, the assumption must be dropped and the model refined.

Key Point: In this book, we will always state our modeling assumptions explicitly before solving a problem. You should do the same in your work. Writing 'assume frictionless' or 'neglect air resistance' is not sloppy physics — it is careful physics that signals to the reader exactly what you are and are not accounting for.

5.10 Experimental Error and Uncertainty

No measurement is perfectly precise. Every measured value differs from the true value by some amount, due to the limitations of the measuring instrument, the skill of the experimenter, environmental variability, and the fundamental quantum nature of matter at very small scales. This unavoidable imperfection is called experimental error, though the word “error” does not imply a mistake; it means the natural spread of results.

Uncertainty is the quantitative expression of this spread. A measurement reported as 4.75 ± 0.05 m tells you that the true value is believed to lie between 4.70 m and 4.80 m. The ± 0.05 m is the uncertainty. Reporting uncertainty is not optional in quantitative science; an answer without uncertainty is incomplete.

$$\text{measurement} = \text{best estimate} \pm \text{uncertainty} \quad (\text{unit})$$

Example: $m = 12.4 \text{ g} \pm 0.1 \text{ g}$ means the true mass is between 12.3 g and 12.5 g.

The uncertainty is typically estimated in one of three ways. For a single measurement: use half the resolution of the instrument. For repeated measurements: use half the range (max – min) or one standard deviation. For calculated results: use propagation of uncertainty (Section 5.12).

5.11 Random Error vs. Systematic Error

Not all errors are alike. Understanding the difference between random and systematic errors is essential for improving experimental results.

Random errors are unpredictable, non-repeating fluctuations. They cause repeated measurements to scatter around the true value. Random error affects precision but not accuracy: the average of many measurements tends toward the true value, because positive and negative fluctuations cancel. Random error is reduced by taking more measurements and averaging them.

Systematic errors are consistent, repeatable deviations in one direction. They shift every measurement by the same amount, so averaging does not help. Systematic errors affect accuracy. They arise from miscalibrated instruments, parallax in reading a scale, poorly defined procedures, or incorrect assumptions about the measurement system. Identifying and eliminating systematic error requires careful experimental design, not just more data.

Feature	Random Error	Systematic Error
Nature	Unpredictable, scattered	Consistent, repeatable

Effect on repeated measurements	Scatter around true value	Consistently above or below true value
Affects accuracy?	No (only precision)	Yes
Reduced by averaging?	Yes	No
Cause	Instrument resolution, human variability	Miscalibration, parallax, flawed procedure
Remedy	Take more measurements; average	Calibrate instruments; redesign experiment

Table 5.7 Comparison of random and systematic errors.

Example — Random error Timing a pendulum ten times with a stopwatch gives values scattered between 1.98 s and 2.04 s. These fluctuations are random; average them to get 2.01 s.

Example — Systematic error A ruler whose first millimeter has worn away always reads 1 mm too small. Measuring the same rod gives 14.83 cm every time — precise, but systematically wrong.

5.12 Propagation of Uncertainty (Introductory)

When a calculated quantity depends on two or more measured quantities, each with its own uncertainty, those uncertainties combine to produce an uncertainty in the result. This process is called propagation of uncertainty (or propagation of error).

For introductory purposes, we use simplified rules that give a reasonable upper estimate of the uncertainty in the result. More precise statistical treatment is covered in the advanced series.

Rule 1: Addition and Subtraction

When adding or subtracting quantities, add the absolute uncertainties.

$$\text{If } z = x + y \text{ or } z = x - y, \text{ then } \Delta z = \Delta x + \Delta y$$

Example — Adding two lengths $L_1 = 3.50 \pm 0.05 \text{ m}$; $L_2 = 2.20 \pm 0.03 \text{ m}$; $L_{\text{total}} = 5.70 \pm 0.08 \text{ m}$

Rule 2: Multiplication and Division

When multiplying or dividing quantities, add the relative (percentage) uncertainties.

$$\text{If } z = xy \text{ or } z = x/y, \text{ then } \Delta z/z = \Delta x/x + \Delta y/y$$

Example — Calculating area $L = 4.0 \pm 0.1 \text{ m}$ (2.5%); $W = 2.5 \pm 0.1 \text{ m}$ (4.0%); Area $= 10 \text{ m}^2 \pm (2.5+4.0)\% = 10 \text{ m}^2 \pm 0.65 \text{ m}^2 \approx 10 \pm 0.7 \text{ m}^2$

Rule 3: Power

When raising a quantity to a power n , multiply the relative uncertainty by $|n|$.

$$\text{If } z = x^n, \text{ then } \Delta z/z = |n| \times \Delta x/x$$

Example — Squaring a measurement $r = 5.0 \pm 0.2 \text{ cm}$ (4.0%); $r^2 = 25 \text{ cm}^2 \pm (2 \times 4.0\%) = 25 \pm 2.0 \text{ cm}^2$

Key Point: These rules give the maximum possible uncertainty in the result. In practice, because errors can partially cancel, the actual uncertainty is often smaller. But these rules are safe and conservative — a good starting point for any lab report.

Bridge Note *The advanced series uses the statistical (quadrature) method for propagating uncertainty: uncertainties are added in quadrature ($\sqrt{(\Delta x)^2 + (\Delta y)^2}$), which gives a more accurate estimate when errors are independent and random.*

5.13 Accuracy, Precision, and Reliability in Experiments

A good experiment must meet three criteria: it must be accurate (close to the true value), precise (reproducible), and reliable (robust against changes in conditions, equipment, and experimenter).

Accuracy is checked by comparing to an accepted or standard value. Percent error measures accuracy:

$$\text{percent error} = |\text{measured} - \text{accepted}| / \text{accepted} \times 100\%$$

Precision is measured by the spread of repeated measurements. A small spread (small uncertainty) indicates high precision. Precision is limited by the resolution of the instrument and the consistency of the experimental procedure.

Reliability refers to whether the result holds up when others repeat the experiment with different equipment, in different laboratories, and under different conditions. A result that only works in one lab, with one instrument, on one day is not reliable. Scientific knowledge is built from results that are confirmed across independent repetitions.

Criterion	Question Asked	How Measured / Assessed	Improved By
Accuracy	Is the result close to truth?	Percent error vs. accepted value	Calibrating instruments; eliminating

			systematic error
Precision	Are results reproducible?	Uncertainty; spread of repeated measurements	Better instruments; cleaner procedure; more trials
Reliability	Do others get the same result?	Independent replication	Clear procedure; well-controlled variables; peer review

Table 5.8 *The three criteria of experimental quality and how each is assessed and improved.*

5.14 Common Mistakes in Data Interpretation

Mistake 1: Connecting data points with a zigzag line

Individual data points scatter due to random error. A zigzag connecting every pair of adjacent points captures that scatter rather than the underlying trend. Draw a smooth best-fit line or curve instead.

Mistake 2: Forcing the best-fit line through the origin

The line of best fit should not be forced through the origin unless theory specifically requires a zero-zero relationship. If the data's best-fit line has a non-zero y-intercept, that intercept carries physical information — perhaps an initial value, an offset, or a systematic error.

Mistake 3: Reading slope from data points instead of the best-fit line

Slope must be calculated from two widely separated points on the best-fit line, not from raw data points. Data points scatter; the best-fit line averages through the scatter.

Mistake 4: Extrapolating beyond the data range

A linear relationship observed between $x = 0$ and $x = 10$ may become nonlinear at $x = 15$. Extrapolating without theoretical justification produces unreliable results. Always label extrapolated values as such.

Mistake 5: Ignoring units on graph axes

A graph without units on its axes is meaningless. The slope's units are y-units/x-units. A graph labeled 'Position vs. Time' with no units tells you nothing about the speed; 'Position (m) vs. Time (s)' tells you the slope will be in m/s.

Mistake 6: Reporting more significant figures than the uncertainty supports

If a length has uncertainty ± 0.1 m, reporting it as 3.1476 m is meaningless. The last three decimal places are below the detection limit of the measurement. Round to 3.1 m.

± 0.1 m.

Mistake 7: Ignoring uncertainty entirely

A numerical result without an uncertainty estimate cannot be compared to theory or to other measurements. Without uncertainty, you cannot assess whether agreement is real or coincidental. Always report uncertainty.

Common Mistake: The most dangerous graph error is the 'ruler through any two convenient points' method for finding slope. Choose the two points widely separated on the best-fit LINE — not two raw data points — and the slope will be much more reliable.

Chapter Summary

Graphs, models, and uncertainty analysis are the analytical infrastructure of experimental physics. Together, they allow physicists to extract relationships from data, simplify complex systems, and report results with appropriate honesty about their limitations.

- The independent variable is plotted on the x-axis; the dependent variable on the y-axis. All other variables are controlled.
- A scientific graph requires: title, labeled axes with units, data points, and a best-fit line or curve.
- Slope = $\Delta y / \Delta x$, calculated from two points on the best-fit line. Its units and physical meaning depend on what is plotted.
- Area under a graph = (y-units) $\times$ (x-units); it represents an accumulated quantity.
- Common curve shapes: linear ($y = mx + b$), parabolic ($y = kx^2$), inverse ($y = k/x$), exponential. Linearization transforms nonlinear data into a straight line for easier analysis.
- A model is a deliberate simplification that retains essential physics and ignores secondary effects. Models must state their assumptions and their domain of validity.
- Random error causes scatter around the true value; it is reduced by averaging. Systematic error shifts all readings in one direction; it is eliminated by calibration and better experimental design.
- Report measurements as value $\pm$ uncertainty. Uncertainty propagates through calculations: add absolute uncertainties for addition/subtraction; add relative uncertainties for multiplication/division.

- Accuracy (closeness to truth) is measured by percent error. Precision (reproducibility) is measured by the spread of repeated readings. Reliability requires independent replication.

Key Terms

independent variable: The variable the experimenter controls or varies; plotted on the horizontal (x) axis.

dependent variable: The variable that responds to changes in the independent variable; plotted on the vertical (y) axis.

controlled variable: A quantity held constant throughout an experiment to prevent it from affecting the dependent variable.

best-fit line: A straight line drawn through a scatter of data points that minimizes the total deviation of the points from the line; represents the underlying trend.

slope: The ratio $\Delta y/\Delta x$ for a straight line; represents the rate of change of the dependent variable with respect to the independent variable.

y-intercept: The value of y when $x = 0$; the point where the best-fit line crosses the vertical axis.

area under the curve: The area enclosed between a graph and the horizontal axis; represents an accumulated (integrated) quantity.

interpolation: Estimating the value of a variable between two measured data points.

extrapolation: Estimating the value of a variable outside the range of the data; unreliable without theoretical justification.

linearization: Replotting data (e.g., y vs. x^2 instead of y vs. x) to convert a nonlinear relationship into a straight line.

physical model: A deliberate simplification of a real system that retains essential physics while ignoring secondary effects.

idealization: A specific simplifying assumption within a model (e.g., 'frictionless surface,' 'point mass,' 'ideal gas').

assumption: An explicit statement of a condition taken to be true for the purposes of a model or calculation.

experimental error: The unavoidable discrepancy between a measured value and the true value; does not imply a mistake.

uncertainty: The quantitative estimate of the range within which the true value of a

measurement is believed to lie; expressed as $\pm$ a value.

random error: Unpredictable, non-repeating measurement fluctuations that cause scatter around the true value; reduced by averaging.

systematic error: A consistent, repeatable deviation of measurements from the true value in one direction; not reduced by averaging; requires calibration or redesign.

propagation of uncertainty: The process of calculating the uncertainty in a derived quantity from the uncertainties of the measured quantities used to compute it.

percent error: A measure of accuracy: $|\text{measured} - \text{accepted}| / \text{accepted} \times 100\%$.

precision: The reproducibility of a measurement; measured by the spread of repeated readings.

accuracy: The closeness of a measurement to the true or accepted value.

reliability: The property of an experimental result that holds up under independent replication by other experimenters.

Concept Review Questions

Answer in complete sentences. No calculation required.

1. What is the difference between the independent and dependent variables in an experiment? Give an example from daily life.
2. List six elements a properly constructed scientific graph should include. Why does each matter?
3. Explain why a best-fit line should not be drawn by connecting data points with straight line segments between each pair.
4. What physical quantity does the slope represent on a position-time graph? On a velocity-time graph? On a force-extension graph?
5. What physical quantity does the area under a velocity-time graph represent? Why does the unit analysis support this?
6. Describe two different curve shapes that appear in physics graphs and give one physical example of each.
7. What is a physical model? What is an idealization? Give one example of each from this chapter.
8. Why is stating your assumptions explicitly, before solving a problem, considered good scientific practice?

9. What is the difference between random error and systematic error? Which one is reduced by taking more measurements and averaging? Explain why.
 10. A student measures the period of a pendulum ten times and gets values ranging from 1.96 s to 2.04 s. Estimate the uncertainty, and report the result in proper format.
 11. What does it mean to say a result is precise but not accurate? What does it mean to say it is accurate but not precise?
 12. Explain two common mistakes students make when reading slope from a graph.
-

Graph Interpretation Exercises

Use the descriptions below to answer each question as if you had the graph in front of you. No graph paper needed; reason from the given values.

13. A position-time graph shows a straight line passing through (0 s, 2.0 m) and (5.0 s, 17.0 m). (a) What is the slope? (b) What are its units? (c) What physical quantity does the slope represent? (d) What does the y-intercept represent physically?
 14. A velocity-time graph shows a straight line from (0 s, 0 m/s) to (8.0 s, 24.0 m/s). (a) What is the slope? (b) What physical quantity is this? (c) What is the area under the line from $t = 0$ to $t = 8.0$ s? (d) What does that area represent?
 15. A force-extension graph shows a straight line through the origin with slope 250 N/m. (a) What is this slope called? (b) What force is needed to extend the spring 0.12 m? (c) If the line becomes curved at extensions beyond 0.30 m, what does this indicate about the spring?
 16. A graph of y vs. x^2 for a free-fall experiment gives a straight line through the origin with slope 4.9 m/s^2 . (a) What is the original relationship between y and x ? (b) What does the slope represent? (c) Why did the experimenter plot y vs. x^2 instead of y vs. x ?
 17. A velocity-time graph for a car shows: $v = 0$ from $t = 0$ to $t = 2$ s; v increases linearly from 0 to 15 m/s between $t = 2$ s and $t = 6$ s; $v = 15$ m/s from $t = 6$ s to $t = 10$ s. (a) Find the acceleration between $t = 2$ s and $t = 6$ s. (b) Find the total distance traveled from $t = 0$ to $t = 10$ s.
-

Graph Construction Practice

For each dataset, construct a properly labeled graph on graph paper (or describe exactly how you would construct it). Determine the best-fit line equation and extract the relevant physical quantity from the slope.

18. A spring is stretched by hanging different masses. Record:

Mass (g)	Extension (cm)
100	3.8
200	7.6
300	11.4
400	15.0
500	18.8

(a) Plot extension (y) vs. mass (x). (b) Draw the best-fit line. (c) Find the slope in units of cm/g. (d) Convert to N/m and identify the spring constant.

19. A ball rolls down a ramp; position is measured every 0.5 s:

Time (s)	Position (m)
0.0	0.00
0.5	0.12
1.0	0.49
1.5	1.08
2.0	1.94

(a) Plot position vs. time. (b) What curve shape do you see? (c) Plot position vs. time². (d) Find the slope in m/s². (e) What does the slope represent physically?

Experimental Uncertainty Problems

Show all work. Report final answers in the form value $\pm$ uncertainty (unit).

20. A block's length is measured five times: 8.23, 8.25, 8.22, 8.26, 8.24 cm. (a) Find the mean. (b) Find the range. (c) Estimate the uncertainty as half the range. (d) Report the result properly.
21. A force $F = 25.0 \pm 0.5$ N acts over displacement $d = 4.0 \pm 0.2$ m. Work $W = Fd$. (a) Find W . (b) Find the percent uncertainty in F . (c) Find the percent uncertainty in d . (d) Find the percent uncertainty in W . (e) Express W with uncertainty.
22. The accepted value of g is 9.81 m/s². A student measures 9.65 m/s². Calculate the percent error.
23. A radius $r = 3.0 \pm 0.1$ cm is used to compute area $A = \pi r^2$. (a) Find A . (b) Find the percent uncertainty in r . (c) Find the percent uncertainty in A (remember to multiply by the power 2). (d) Express A with uncertainty.
24. Two resistors are measured: $R_1 = 120 \pm 3$ Ω and $R_2 = 80 \pm 2$ Ω . They are connected in series ($R_{\text{total}} = R_1 + R_2$). (a) Find R_{total} . (b) Find the uncertainty in R_{total} .
25. A student measures the period of a pendulum five times: 1.41, 1.44, 1.42, 1.45, 1.43 s. The accepted value is 1.41 s. (a) Report the mean with uncertainty. (b) Calculate percent error. (c) Is the accepted value within the student's uncertainty range?

Lab Analysis Activity

Ramp and Roll — Graph and Model Analysis

Purpose

To practice data collection, graph construction, slope extraction, uncertainty estimation, and model evaluation in a single integrated experiment.

Materials

- A smooth ramp (a board propped at one end with a stack of books)
- A small ball or cart
- A ruler (millimeter resolution)
- A stopwatch or phone timer
- Graph paper

Procedure

26. Set the ramp at a fixed angle. Release the ball from the top of the ramp. Measure the time it takes to travel five different fixed distances (e.g., 20, 40, 60, 80, 100 cm from the release point). Repeat each measurement three times and take the average.
27. Construct a data table with columns: Distance (cm), Trial 1 Time (s), Trial 2 Time (s), Trial 3 Time (s), Mean Time (s), Uncertainty (s).
28. Plot Distance vs. Mean Time. Describe the curve shape.
29. Plot Distance vs. Mean Time². Check whether this gives a straight line. If so, draw the best-fit line and calculate its slope in cm/s².
30. The model for constant acceleration from rest is $d = \frac{1}{2}at^2$. Compare your slope to $\frac{1}{2}a$. What is the acceleration down the ramp?

Analysis Questions

31. What idealizations does the model $d = \frac{1}{2}at^2$ make? List at least two.
32. If your graph of d vs. t^2 did not produce a perfectly straight line, what does that suggest about the model?
33. Identify one source of random error and one source of systematic error in your experiment. How could each be reduced?
34. Calculate the percent uncertainty in your slope by taking the range of slopes from the extreme points on the best-fit line.
35. If you doubled the angle of the ramp, what do you predict would happen to the slope of your d vs. t^2 graph? Explain using the model.

Chapter 5 Checkpoint

Before moving on to Chapter 6, confirm you can do each of the following without notes.

I can...	Section
Identify the independent and dependent variables in an experiment	5.2
List the six required elements of a properly constructed scientific graph	5.4

Calculate slope from a graph using two points on the best-fit line	5.5
State the physical meaning of slope for position-time, velocity-time, and force-extension graphs	5.5
Explain what the area under a velocity-time graph represents	5.6
Identify whether a graph shows a linear, parabolic, inverse, or exponential relationship	5.7
Linearize a simple nonlinear dataset by replotting with a transformed variable	5.7
Define and give an example of a physical model and an idealization	5.8–5.9
Distinguish random error from systematic error and state how each is reduced	5.11
Report a measurement with its uncertainty: value $\pm \Delta$ (unit)	5.10
Propagate uncertainty through addition/subtraction and multiplication/division	5.12
Calculate percent error given a measured and accepted value	5.13
Identify seven common data interpretation mistakes	5.14

Continue to Chapter 6: Describing Motion →